Combining Airborne LiDAR Data and Optical Imagery for Improved National-Scale Beach Topography Estimation: A Case Study in New Zealand

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***Abstract*—Accurate beach topography mapping is crucial for understanding coastal dynamics and mitigating climate change impacts. However, traditional methods such as airborne LiDAR have limitations, leading to substantial gaps in national-scale elevation data. This study presents an innovative framework to reconstruct missing elevation data along New Zealand's (NZL) coastline by integrating airborne LiDAR, Sentinel-2 optical imagery, and geometric features (distance) using machine learning (ML) methods. Our results show that artificial neural network (ANN) emerged as the best model (test set: R^2 = 0.79, root mean squared error (RMSE) = 0.91 m; validation set: 0.79, RMSE = 0.93 m), outperforming other models in accuracy. The produced 10-m digital elevation model (DEM) for national-scale sandy beaches expands area coverage by 286.6% (114.15 km^2), filling gaps in 1249 beaches, including remote areas such as Stewart Island. This novel framework offers a scalable solution for improving the comprehensiveness and accuracy of beach topography. It provides essential support for inundation prediction, habitat management, and the development of climate adaptation strategies, thereby facilitating more informed decision-making in coastal zone management and climate change mitigation efforts.**

***Index Terms*—Airborne LiDAR, beach, coastal, digital elevation model (DEM), machine learning (ML).**

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I. Introduction

BEACHES, defined as sedimentologically active transition zones between terrestrial and marine systems, which comprise unconsolidated deposits spanning three orders of magnitude in grain size—from sand (0.0625–2 mm) through gravel (2–64 mm) to cobble (>64 mm) [1], represent dynamic interfaces between terrestrial and marine environments. Shaped by the dynamic interplay of wave action, tidal forces, and aeolian processes, beach morphology is modulated by coastal geomorphology, sediment composition, and hydrodynamic conditions [2], [3]. As low-lying coastal landforms with elevations generally below 10 m, beaches are highly susceptible to climate change-induced hazards, including coastal erosion, storm surges, and sea-level rise, underscoring their importance in coastal management strategies and hazard risk assessments for sustainable shoreline preservation [4], [5], [6].

Beach topography serves as a fundamental control on coastal dynamics, with its 3-D morphology and elevation distribution influencing coastal evolution through multiple mechanisms. First, topographic features including slope gradient, dune height, and sandbar morphology directly determine wave energy dissipation efficiency, sediment transport pathways, and shoreline stability [7], [8], [9]. Second, intertidal topography functions as a critical transitional zone, dissipating wave energy through berm erosion during storms while facilitating onshore sediment transport and dune recovery

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during calm periods [7], [10]. Furthermore, topographic characteristics modulate beach morphodynamic states (e.g., reflective or dissipative) by altering nearshore hydrodynamics, thereby controlling the system's resilience to extreme events and sea-level rise [11]. Advanced monitoring techniques (e.g., LiDAR and UAV photogrammetry) provide essential data for quantifying these processes [12], while underlying geology and anthropogenic modifications continuously reshape topographic patterns and coastal resilience [11].

High-resolution digital elevation models (DEMs) derived from remote sensing provide valuable insights into coastal spatiotemporal dynamics [9], facilitating more accurate predictions of beach erosion, sediment transport, and dune morphology changes [13]. Accurate DEMs are essential for monitoring beach morphodynamics, assessing coastal vulnerability, and informing disaster mitigation strategies [14], [15]. Given the dynamic nature of coastal environments, traditional in situ monitoring often fails to capture rapid changes across large areas [16]. Consequently, remote sensing—particularly airborne LiDAR and Sentinel-2 optical imagery—has become the cornerstone for generating high-resolution beach DEMs [17].

Airborne LiDAR has transformed coastal monitoring through the generation of DEMs with subdecimeter vertical accuracy [18], proving highly effective in quantifying dune morphodynamics and shoreline evolution [19]. Nevertheless, three primary limitations restrict its applications in comprehensive beach topography mapping: 1) tidal fluctuations result in substantial data missing since LiDAR is unable to penetrate water surfaces, leading to elevation data gaps for submerged and intertidal zones [20]; 2) incomplete spatial coverage of exposed beach areas often arises from constraints in flight line spacing and survey timing [21]; and 3) high costs make repeated and large-scale surveys financially unsustainable [22], [23]. Moreover, LiDAR's sensitivity to adverse weather conditions and dependence on specialized equipment hinder its ability in large-scale coastal monitoring programs [24], [25], [26].

Optical satellite imagery has become a cornerstone for coastal topography inversion due to its ability to provide extensive spatiotemporal coverage, making it ideal for large-scale coastal monitoring [9], [13], [24], [27]. For example, time-series imagery integrated with percentile composites has effectively captured intertidal topography and monitored morphological changes across different tidal phases [17]. Such imagery allows for the delineation of key coastal features such as sandy beaches, vegetated dunes, and intertidal zones, thus facilitating the analysis of shoreline dynamics and sediment transport processes [28]. Moreover, optical imagery has been successfully applied in mudflat intertidal zone topography inversion, such as through the use of frequency-based methods, which improve topographic accuracy in muddy areas [17], [20]. However, while significant progress has been made in related fields [17], [29], applying optical imagery specifically for beach topography inversion remains largely unexplored. This gap limits the ability to obtain complete and accurate beach elevation data, particularly in dynamic and complex coastal environments [2], [30]. Currently, there is a lack of systematic research on the combination of optical imagery and sandy beach morphology analysis, leading to an urgent need to develop new methods to explore the potential of remote sensing in acquiring comprehensive sandy beach topography [31].

In recent years, numerous studies have demonstrated that machine learning (ML) methods hold substantial potential for mapping coastal zones, sandy beaches, intertidal areas, and estimating nearshore bathymetry at various spatial scales. Compared with traditional remote sensing and field survey approaches, ML techniques exhibit notable advantages in improving inversion accuracy, enabling automated processing, and facilitating large-scale applications. Various algorithms—including support vector machines (SVMs), random forests (RFs), neural networks (NNs), and deep learning architectures such as U-Net and long short-term memory (LSTM)—have been successfully applied to elevation and bathymetry inversion for coastal, beach, and intertidal environments, achieving significant improvements in both accuracy and automation levels [32], [33], [34], [35], [36], [37]. Furthermore, ML has been integrated not only with conventional optical remote sensing imagery but also with diverse data sources such as LiDAR (e.g., ICESat-2), multitemporal satellite imagery, and UAV surveys, thereby enhancing its applicability in complex environmental conditions [32], [33], [34], [36]. Collectively, these advances confirm the feasibility and growing importance of ML-based approaches for high-resolution coastal topographic mapping and provide a strong foundation for the methodology proposed in this study.

Despite advances in remote sensing and terrain modeling technologies, widely used coastal DEMs—such as the DeltaDTM [38], the Coastal National Elevation Database (CoNED) [39], and CoastalDEM [40]—remain inadequate for representing detailed beach topography. These datasets are commonly limited by relatively low spatial resolution (e.g., DeltaDTM at 30-m resolution, derived from CopernicusDEM, and CoastalDEM at 30-m resolution, based on SRTM) and decreased accuracy in complex intertidal zones, making them insufficient for capturing the fine-scale geomorphic features of beaches. Moreover, beach areas are often characterized by data gaps, discontinuities, or significant vertical errors in these DEMs, which undermines their utility in accurately monitoring and modeling beach dynamics. As a result, these limitations hinder a wide range of critical applications, including beach vulnerability assessments, sea-level rise inundation modeling, coastal erosion analysis, and the sustainable management and conservation of beach environments. Currently, the lack of a high-resolution, beach-specific terrain database with integrated land-sea coverage remains a major constraint for advancing research and decision-making related to beach systems.

To address these limitations, this study introduces an innovative methodology for national-scale beach topographic mapping by integrating airborne LiDAR, Sentinel-2 optical imagery, and coastal morphological geometric features. Our approach introduces a novel fusion framework that enables accurate elevation estimation for beaches lacking LiDAR data, improving coastal DEM generation. We have developed New Zealand's (NZL) first national-scale beach topography dataset,

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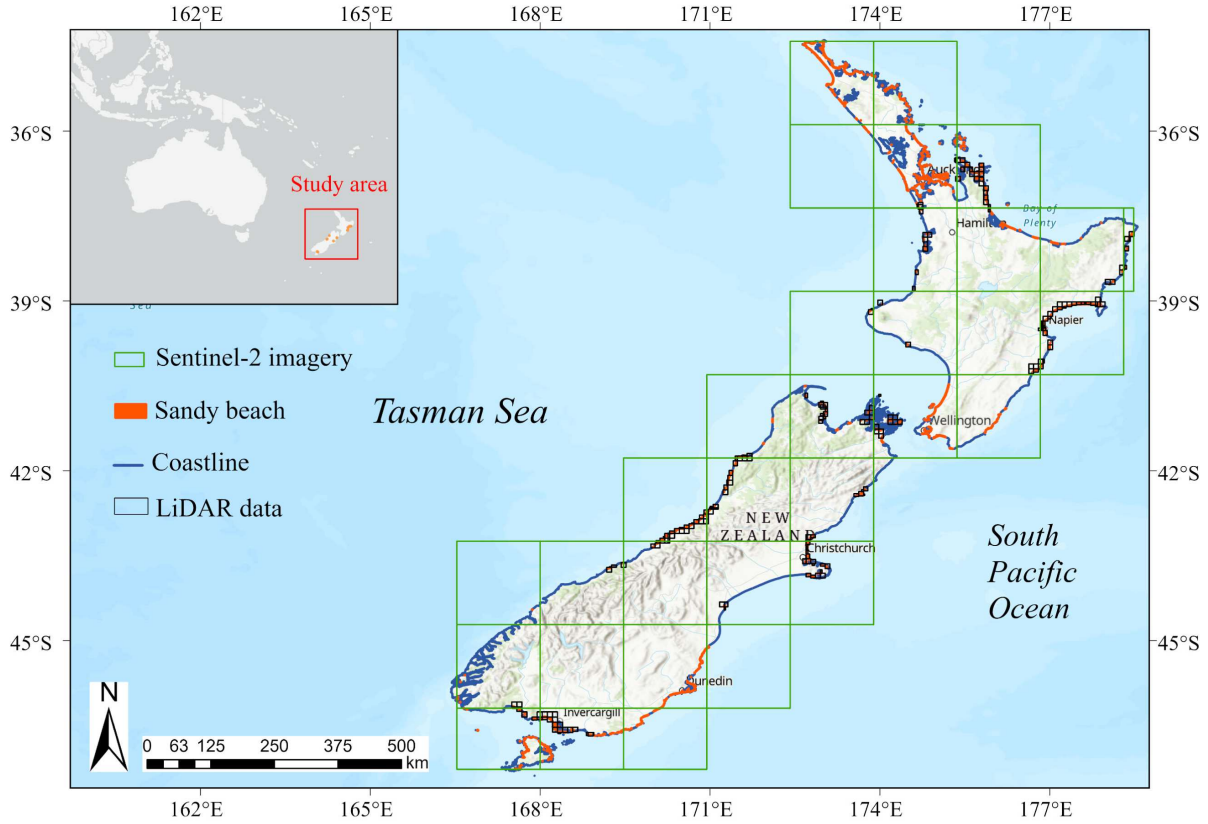


Fig. 1. Map of the study area (NZL). The green boxes represent the locations of 31 Sentinel-2 image scenes. The red areas indicate beaches (a total of 1608 beach polygons). The blue line marks the coastline of NZL, and the black boxes show the coverage of 183 LiDAR-derived DEM tiles. In the inset map at the top-left corner, the red area highlights the geographical location of our study region.

offering a crucial and reliable foundation for effective coastal zone management.

II. Study Areas and Materials

A. Study Area

The study area covers the extensive coastal zones of NZL, encompassing both the North and South Islands as well as other major islands, including Stewart Island, the Chatham Islands, and Great Barrier Island [41]. NZL has approximately 15 000 km of coastline, making it one of the longest national coastlines in the world relative to land area [42]. This coastline includes various beach types, including sandy beaches, gravel-dominated shorelines, and mixed sediment beaches [43]. These coastal environments are influenced by the surrounding Pacific Ocean to the east and the Tasman Sea to the west, which shape the hydrodynamic and sedimentary processes that govern beach morphology [44]. NZL's coastal zones are highly dynamic and vulnerable to climate-induced changes, with sea level rising at an average rate of 3.9 mm/year over the past two decades, which is slightly higher than the global average of 3.4 mm/year [45], [46]. The geographical location and distribution of data in NZL are shown in Fig. 1 (including 1608 beaches).

B. Airborne LiDAR Data

The airborne LiDAR data were sourced from land information NZL (https://data.linz.govt.nz/layer/121859-new-zealand-lidar-1m-dem/). These data were employed to generate a high-resolution DEM with a spatial resolution of 1 m via aerial surveying. The dataset consists of 183 tiles in raster TIFF format, spanning the time period from 2020 to 2024. The original DEM data are referenced to the NZL Transverse Mercator 2000 (NZTM2000) coordinate system, and the vertical datum is set to the NZL Vertical Datum 2016 (NZVD2016). This configuration ensures a vertical accuracy of ± 0.2 m, which is crucial for precise geospatial analysis in the study of NZL's coastal zones. The basic information of the LiDAR DEM imagery is presented in Table I, which includes some fundamental details of image acquisition and vertical accuracy.

We calculated the minimum coastal elevation of these 183 DEMs at their acquisition times, which theoretically represents the tidal levels at the time of image capture for these 183 scenes. The distribution of these tidal level values is shown in Fig. 2(a). In addition, we computed the coverage ratio of the DEMs across 1608 sandy beaches. The distribution of the coverage ratios is presented in Fig. 2(b).

C. Sentinel-2 Optical Imagery

In this study, the Sentinel-2 (including Sentinel-2A and Sentinel-2B) data were among the main data sources. The Sentinel-2 satellite has a revisit cycle of 5 days at the equator, allowing for frequent and high-resolution observations. Sentinel-2 imagery has four 10-m resolution bands, including blue (B2: 490 nm), green (B3: 560 nm), red (B4: 665 nm), and

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TABLE I
BASIC INFORMATION OF LIDAR DEM IMAGE

Acquisition time	Vertical datum	Vertical accuracy	Number of images	Spatial resolution
2020-2024	NZVD2016	±0.2 m	183 tiles	1 m

TABLE II
SENINEL-2 IMAGE BASIC INFORMATION

Acquisition time	Percentage composition type	Data type	Bands	Spatial resolution
2020-2024	20th, 50th and 80th percentile	Level-2A	B2, B3, B4 and B8	10 m

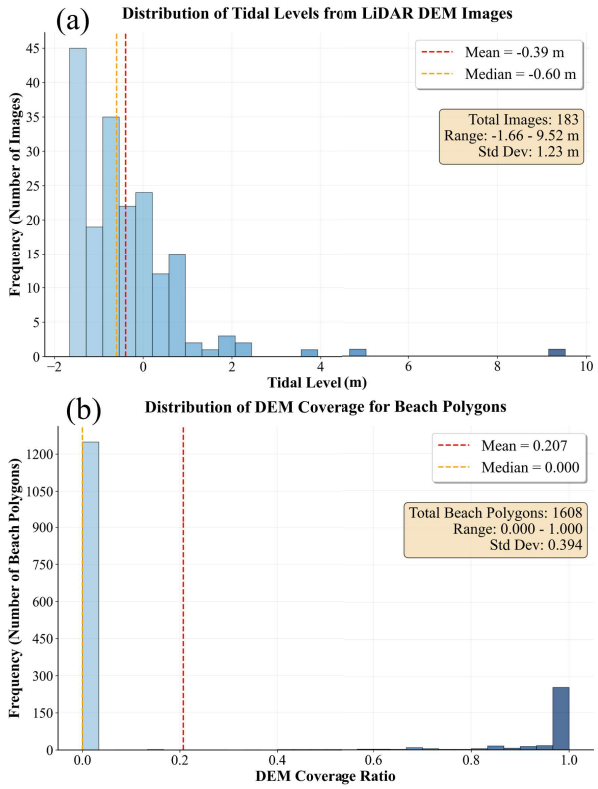


Fig. 2. Histograms of LiDAR DEM data analysis. (a) Frequency histogram of tidal levels derived from LiDAR DEM data. (b) Frequency histogram of coverage ratios of LiDAR DEM data over sandy beaches. In the histograms, darker shades indicate higher values.

near-infrared (B8: 842 nm). Here, the Sentinel-2 data were obtained via Google Earth Engine (GEE) [47]. We acquired all available Sentinel-2 imagery from January 1, 2020, to December 31, 2024, utilizing the Level-2A surface reflectance product (COPERNICUS/S2_SR_HARMONIZED). The Level-2A product has been processed with atmospheric correction using the Sen2Cor algorithm [48], which removes atmospheric effects to provide bottom of atmosphere (BOA) reflectance. In addition, cloud masking was applied to exclude contaminated pixels and ensure data quality, enhancing the reliability of the acquired time-series imagery for further analysis. The detailed information on Sentinel-2 imagery is shown in Table II, which includes information such as acquisition time, percentile composition type, data type, bands, and spatial resolution.

D. Auxiliary Data

Moreover, to accurately define the boundaries and geometric characteristics of NZL's beaches, sandy beaches, and coastline data in shapefile format sourced from OpenStreetMap (OSM) were applied. The OSM-derived coastline data provided the baseline for computing key beach geometric parameters (e.g., distance-to-shoreline), while the 1608 identified beaches formed the essential spatial framework for subsequent analysis.

III. METHODOLOGY

The comprehensive workflow for mapping beach topography at the national scale comprises four key steps: 1) data preprocessing; 2) feature engineering; 3) model selection, optimization, and interpretation; and 4) national scale DEM generation. The workflow is shown in Fig. 3.

A. Data Preprocessing

This section involves the preprocessing of LiDAR DEM data and optical remote sensing imagery for subsequent analysis.

Regarding the LiDAR DEM data preprocessing, it involves four key steps.

1) *Geodetic Transformation:* The DEM rasters are transformed into the WGS-84 coordinate system to maintain geodetic consistency with the optical images.
2) *Resampling:* Utilizing the nearest neighbor method, the DEM imagery is resampled to a 10-m resolution, ensuring alignment with the pixel grid of Sentinel-2 imagery.
3) *Water Masking:* For each LiDAR scene covering coastal sandy beaches, we applied a frequency-based water identification approach to remove sea surfaces [49], [50]. Leveraging the principle that LiDAR-derived sea surface elevations are consistent across each scene, we calculated the elevation frequency distribution. For each scene, the peak frequency value was identified to reflect sea surfaces, and only pixels with elevations greater than this threshold were included in further analysis.

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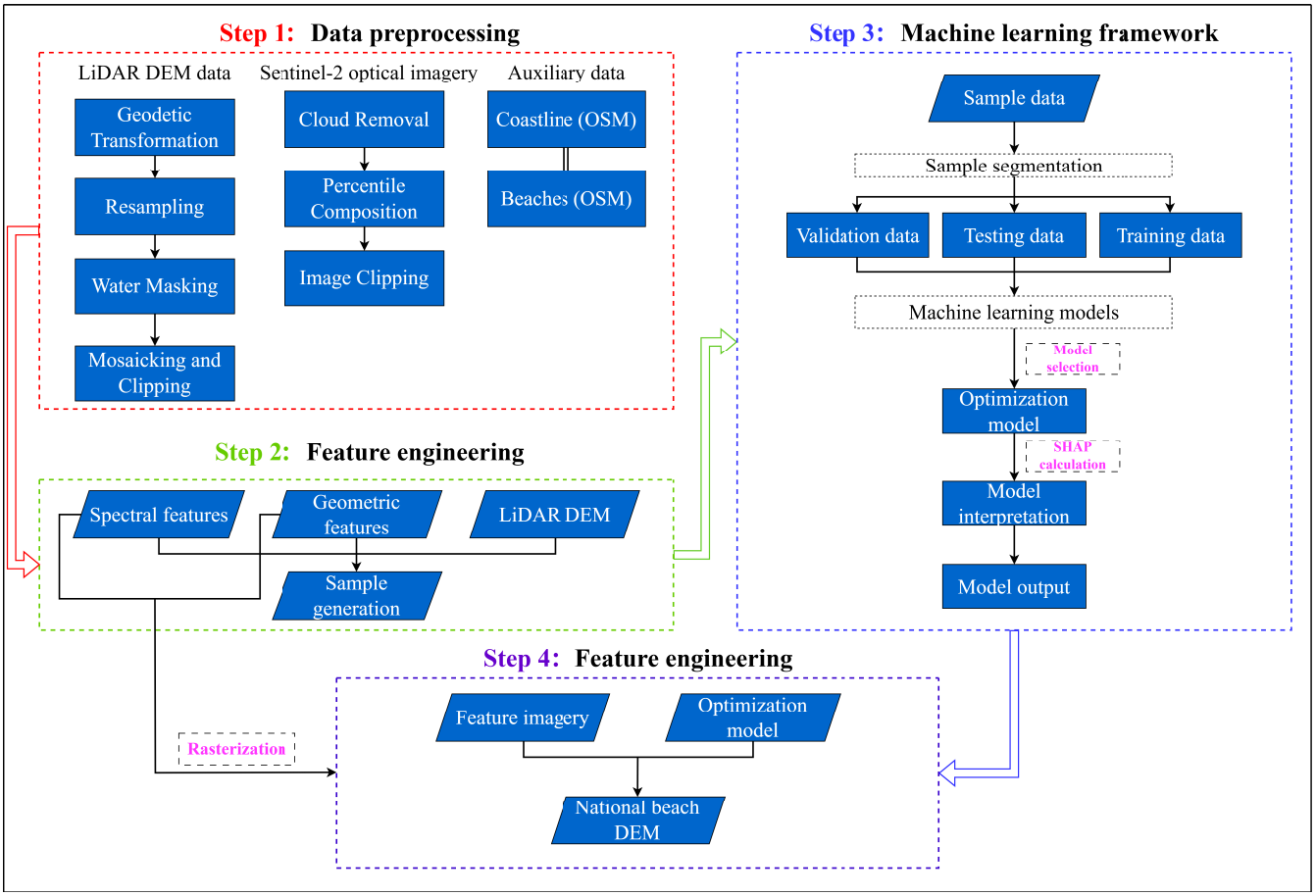


Fig. 3. Workflow for nation-scale beach topography mapping in NZL.

For validation, we selected two representative scenes from distinct peninsula regions in NZL (Canterbury and Hawke's Bay). As summarized in Fig. 4, the results include: (g) geographical locations of the two scenes, (a) and (b) 3-D topographic maps, (c) and (d) elevation histograms, and (e) and (f) cross-sectional profiles along selected transects. The analysis reveals that sea surfaces exhibit exceptionally prominent frequency peaks [see Fig. 4(a)–(f)]. Notably, even in topographically flat beach areas (e.g., Canterbury region in Fig. 4), the frequency signal associated with seawater remains clearly dominant. Fig. 4 further demonstrates that despite minor elevation variations across flat beaches, sea surfaces maintain consistent elevation.

4) *Image Mosaicking and Clipping:* All resampled DEMs are mosaicked into a single image. Subsequently, based on the mosaicked image and the beach data (with the shapefile format), the DEM is clipped to the extent of the beaches.

The entire DEM preprocessing workflow was systematically implemented in batch mode through Python 3.11, utilizing the arcpy geoprocessing library embedded in ArcGIS Pro 3.4 software environment.

The preprocessing of optical remote sensing imagery, it involved three steps as follows.

1) Cloud Removal: Clouds and cloud shadows were identified and removed using Sentinel-2's QA60 bands, which provide a reliable mask for atmospheric disturbances.

2) Percentile Composition: To mitigate the effects of cloud contamination and seasonal variability, a time series analysis was conducted using 10-m resolution bands from 2020 to 2024. For each pixel, the 20th, 50th, and 80th percentiles were computed across the time series. These percentiles were used to generate composite images, capturing the lower, median, and upper spectral reflectance values, respectively. This approach effectively reduced noise while retaining the spectral characteristics under different tidal conditions [51], [52].

3) Image Clipping: The percentile composite images were spatially clipped using the beach data.

In this process, the first two steps were implemented on GEE using the updateMask function and percentile function, and the third step was conducted using the clip tool from arcpy's data management module.

Through these preprocessing steps, the dataset was refined to improve both the accuracy and reliability of subsequent analysis.

B. Feature Engineering

1) Spectral Features: Spectral features were extracted from the blue, green, red, and near-infrared bands of the three percentiles (the 20th, 50th, and 80th percentiles) composite images derived from Sentinel-2 imagery. These bands were

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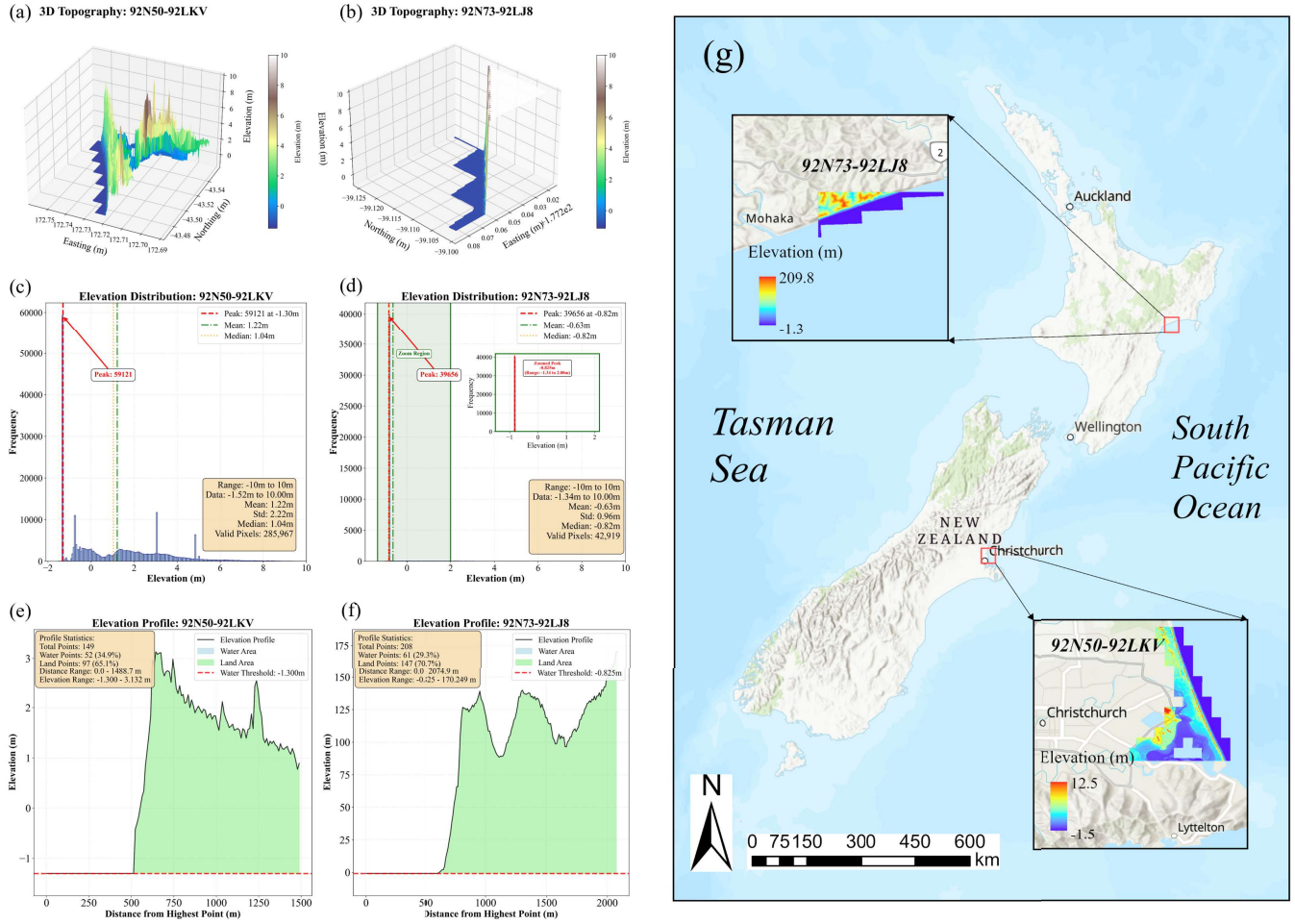


Fig. 4. (a) and (b) Three-dimensional topographic maps (<10 m), (c) and (d) elevation histograms (<10 m), and (e) and (f) distance-varying elevation profiles of the two coastal scenes. (g) Geographical locations of the two selected scenes. (a), (b), and (e) Correspond to the Canterbury region, while (c), (d), and (f) represent the Hawkes Bay region.

chosen because of their high 10-m spatial resolution and unique spectral information, enabling detailed characterization of sandy beaches. Among the four Sentinel-2 bands, the blue band demonstrated heightened sensitivity to beach environments due to its unique ability to penetrate shallow waters, enabling the detection of submerged sandbars and intertidal zones through turbidity-dependent reflectance variations [53]. The near-infrared band could enhance the characterization of beaches by delineating land-water boundaries and moisture gradients, as dry sands exhibited high reflectance while wet or submerged areas absorbed strongly in the near-infrared band. By leveraging various spectral properties of these bands, the extracted features provided a robust foundation for characterizing sandy beaches [54].

2) Geometric Features: For each beach, we quantified a single pixel-level geometric feature: the perpendicular Euclidean distance from the center of each raster pixel to the nearest coastline (distance to coastline, DC, in meters). This feature varies spatially within the beach polygon and provides direct positional context for each pixel relative to the shoreline. The DC was calculated as

$$\mathrm{DC} = \min \sqrt{(x_p - x_c)^2 + (y_p - y_c)^2} \tag{1}$$

where (x_p, y_p) denotes the coordinates of the pixel center and (x_c, y_c) represents the coordinates of coastline vertices.

Table III presents the features utilized for training and their respective sources and the number of features.

3) Sample Set Generation: To prepare the dataset for ML model training, raster images of all selected features (four band images with 20%, 50%, and 80% percentiles and the distance-to-coastline feature) and the preprocessed DEM were converted into point vectors at the pixel centers within areas covered by LiDAR DEM data.

We randomly selected 50% of these pixels as the development set for model-related processes. Within this development set, 70% was used for training, 15% for Bayesian optimization root mean squared error (RMSE) validation and early stopping (except for RF, which cannot use early stopping), and 15% for testing. The remaining 50% of pixels were set aside, from which 20% (10% of total pixels) were randomly sampled as an independent validation set for post-training model evaluation.

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TABLE III
FEATURES USED FOR TRAINING AND THEIR DATA SOURCES

Data Sources	Features	Number of features
Sentinel-2	The 20th, 50th, and 80th percentiles of the blue, green, red, and near-infrared (NIR) bands	12 features
Coastlines (OSM)	The perpendicular distance from pixel center to coastline (DC)	1 featrues

TABLE IV
DATASET PARTITIONING SCHEME

Subset	Percentage of Total Pixels	Purpose
Training	35%	Model training
Bayesian optimization validation & early stopping	7.5%	Hyperparameter tuning and early stopping (except RF)
Testing	7.5%	Stability & generalization testing
Independent validation	10%	Post-training independent evaluation
Hold-out (unused)	40%	Not used

The allocation scheme is summarized in Table IV. Only elevations between −2 and 10 m were included in the analysis.

C. ML Framework

1) Models: In this study, four widely used ML models were employed to establish the relationship between spectral/geomorphic features and LiDAR-derived elevation values. The purpose of selecting these four models—RF [55], XGBoost (XGB) [56], artificial NN (ANN) [57], and hist gradient boosting regressor (HGBR) [58]—was to represent a diverse spectrum of ML paradigms, including bagging-based ensemble methods, traditional and histogram-based gradient boosting techniques, and deep neural architectures. This diversified model suite enabled a comprehensive assessment of their respective strengths in capturing the relationship between spectral and geomorphic features and beach elevation. Such a comparative framework not only facilitated the identification of the most suitable algorithm for elevation prediction but also provided methodological guidance for future coastal topography modeling research. Specifically, the four selected models were described as follows.

a) XGB: It is a scalable and efficient implementation of gradient boosting designed for speed and performance. It builds trees sequentially, where each new tree corrects the errors of the previous ones, making it highly effective for regression tasks. Key parameters include: *n*_estimators (the number of boosting rounds or trees), max_depth (the maximum depth of each tree), learning_rate (the step size shrinkage used to prevent overfitting), subsample (the fraction of samples used for fitting each tree), colsample_bytree (the fraction of features used for fitting each tree), lambda (L2 regularization term on weights), and alpha (L1 regularization term on weights).

b) RF: It is a bagging-based ensemble learning algorithm that builds multiple decision trees independently using bootstrapped subsets of the training data and then aggregates their predictions through averaging for regression tasks. By introducing randomness both in the data sampling and feature selection process, RF effectively reduces variance and enhances model generalization. It is particularly robust against overfitting and performs well with high-dimensional or noisy datasets. Key parameters include: *n*_estimators (the number of trees in the forest), max_depth (the maximum depth of each tree), min_samples_split (the minimum number of samples required to split an internal node), min_samples_leaf (the minimum number of samples required to be at a leaf node), and max_features (the number of features to consider when looking for the best split).

c) ANN: It is a multilayer perceptron (MLP) regression model implemented using the MLPRegressor library. It consists of an input layer, one or more hidden layers, and an output layer, capable of capturing complex nonlinear relationships in the data. Key parameters include: hidden_layer_sizes (the number of neurons in each hidden layer), activation (the activation function for the hidden layers, such as ReLU or tanh), alpha (L2 regularization term to prevent overfitting), learning_rate_init (the initial learning rate for weight updates), max_iter (the maximum number of iterations for training), early_stopping (whether to stop training when

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validation performance stops improving), and batch_size (the size of mini-batches for stochastic optimization).

d) HGBR: It is a highly efficient and scalable implementation of gradient boosting based on histogram-based algorithms. Instead of evaluating all possible split points for each feature, HGBR bins continuous feature values into discrete buckets (histograms), which significantly reduces computation time and memory usage—making it well-suited for large-scale datasets. Like traditional boosting, it builds trees sequentially, with each tree minimizing the residual error of its predecessor. HGBR is optimized for tabular data and supports native handling of missing values. Key parameters include: max_iter (the number of boosting iterations or trees), max_depth (the maximum depth of individual trees and controlling model complexity), learning_rate (controls the contribution of each tree and helps prevent overfitting), and l2_regularization (applies L2 penalty to leaf weights to improve generalization and reduce overfitting).

Here, to ensure comparability and accuracy among the predictions of different models and to avoid uncertainties arising from varying operating environments, all ML models were implemented using Python. Specifically, the RF, HGBR, and ANN models were conducted using the RandomForestRegressor, HistGradientBoostingRegressor, and MLPRegressor from Python's Scikit-Learn library [59], respectively. The XGB model was implemented using the XGBRegressor from the XGB library [56].

2) Hyperparameter Tuning: Prior to model training, it was necessary to standardize the training dataset, which included feature columns X and the elevation column y. We calculated the mean and standard deviation of the samples in the training set. According to the standardization formula [as shown in (2)], both the feature columns and the elevation column were transformed to achieve a mean of 0 and a standard deviation of 1 [60]. This standardization effectively eliminated discrepancies in dimensions and value ranges among different features, enabling a uniform scale for comparative analysis. The standardization parameters derived from the training set were saved for subsequent application to the test and validation sets

$$x' = \frac{x - \mu}{\sigma} \tag{2}$$

where x represents the original data value, x' denotes the standardized data value, μ is the mean of the training set, and σ is the standard deviation of the training set.

For the training process of each model, we employed Bayesian optimization to select the optimal hyperparameters. Compared to traditional grid search or random search, Bayesian optimization offers several advantages: it is more efficient in exploring high-dimensional hyperparameter spaces, requires fewer evaluations to find near-optimal solutions, and can effectively balance exploration of the search space with exploitation of promising regions [61]. The objective function of Bayesian optimization was defined as the RMSE of a 15% validation subset. The optimization was iterated for 100 trials, with an initial 20 random sampling points, and the acquisition function was set to expected improvement (EI). The optimization procedure was implemented using the scikit-optimize package in Python. The detailed parameter settings for each model during training are as follows.

a) XGB Model: The squared error was used as the loss function. To prevent overfitting, early stopping was applied based on the 15% validation set with a patience of 50 rounds. Training stopped if the validation performance did not improve within 50 rounds.

b) RF Model: Similarly, the squared error was used as the loss function. However, since early stopping cannot be implemented in RF, overfitting was controlled by strengthening regularization and explicitly constraining tree size.

c) ANN Model: The squared error was also used as the loss function. Early stopping was applied with the same configuration as in XGB, using the 15% validation set and a patience of 50 epochs.

d) HGBR Model: The squared error was set as the loss function, and early stopping with 50 rounds of patience was also adopted to prevent overfitting.

Finally, the hyperparameter search ranges and the optimal hyperparameter combinations for all models are summarized in Table V. For each ML model, hyperparameter optimization was performed within a predefined search space.

3) Model Interpretation Methods: To enhance the interpretability of the predictive model, we adopted shapley additive explanations (SHAPs) to analyze the contributions of each feature to the model's output. SHAP is a unified framework grounded in cooperative game theory that attributes the prediction of an ML model to each feature based on the concept of Shapley values [62]. These values quantify the marginal contribution of a feature by considering all possible feature combinations, thereby enabling a consistent and locally accurate explanation of the prediction. SHAP generates both global and local explanations by calculating how individual feature values influence model output across all samples and within each instance, respectively. This allows for a transparent understanding of the feature importance and directionality (positive or negative effect) for each prediction.

The SHAP summary plot visualizes the distribution of SHAP values for all features, showing which features have the most significant average impact across the test dataset. The SHAP dependence plot further illustrates how the model's prediction for a specific feature change with its value, highlighting potential interactions or nonlinear effects. Meanwhile, the SHAP heatmap reveals sample-level patterns by visualizing the variation in SHAP values across samples, helping identify spatial or feature-specific sensitivities within the model [62]. These interpretative tools collectively provide insights into the model's behavior, validate feature relevance, and support domain-based reasoning for topographic pattern interpretation.

In this study, we applied SHAP to interpret the optimal model. We performed SHAP analysis on the optimal model, randomly selecting 10 000 samples from the independent validation set for analysis. The distribution of these 10 000 samples was largely consistent with that of the overall validation set. In the feature names, "[number]p_[band]" denotes the specific band of the synthetic imagery at a given percentile (e.g., 20p_NIR represents the near-infrared band of the 20th percentile synthetic image).

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TABLE V
BAYESIAN OPTIMIZATION SEARCH RANGES AND OPTIMAL PARAMETER COMBINATIONS FOR ALL MODELS ARE PRESENTED BELOW

Model	Search space	Best hyperparameters

XGB	n_estimators: 300–1500;
	max_depth: 3–6;
	learning_rate: 0.01–0.1;
	subsample: 0.5–0.8;
	colsample_bytree: 0.5–0.8;
	reg_alpha: 2.0–8.0;
	reg_lambda: 15.0–40.0;
	gamma: 0.1–1.0;
	min_child_weight: 5–20
		n_estimators=1452;
		max_depth=6;
		learning_rate=0.099;
		subsample=0.553;
		colsample_bytree=0.791;
		reg_alpha=3.39;
		reg_lambda=15.14;
		gamma=0.124;
		min_child_weight=6.81;
		early_stopping=True;patience=50

RF	n_estimators: 500–1500;
	max_depth: 10–20;
	min_samples_split: 20–50;
	min_samples_leaf: 10–30;
	max_features: [sqrt, log2, None]
		n_estimators=1448;
		max_depth=20;
		min_samples_split=20;
		min_samples_leaf=10;
		max_features=None

ANN	hidden_layer1: 64–256;
	hidden_layer2: 32–128;
	hidden_layer3: 16–64;
	alpha: 0.001–0.01 (log-uniform);
	learning_rate_init: 0.001–0.01 (log-uniform);
	max_iter: 300–1000;
	batch_size: 64–256;
	activation: [relu, tanh];
	solver: adam
		hidden_layer_sizes=(256, 128, 64);
		alpha=0.00288;
		learning_rate_init=0.001;
		max_iter: 300;
		batch_size=256;
		activation=relu;
		learning_rate=adaptive;
		early_stopping=True;patience=50

HGBR	max_iter: 300–1500;
	max_depth: 5–10;
	learning_rate: 0.01–0.1;
	min_samples_leaf: 10–30;
	l2_regularization: 1.0–10.0;
	max_bins: 100–255;
	max_leaf_nodes: 10–30
		max_iter=1500;
		max_depth=10;
		learning_rate=0.1;
		min_samples_leaf=30;
		l2_regularization=1.0;
		max_bins=255;
		max_leaf_nodes=30;
		early_stopping=True;patience=50

4) Accuracy Validation: Following model training, we evaluated the performance of each model using four key error metrics: RMSE, coefficient of determination (R^2), Bias, and mean absolute error (MAE). We systematically applied these metrics to the training, testing, and validation datasets to ensure a comprehensive assessment of model accuracy and generalization capability at a national scale in NZL. The computational formulas for these metrics are as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{3}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{4}$$

$$\text{Bias} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i) \tag{5}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{6}$$

where y_i represents the observed elevation, $\hat{y}_i$ denotes the predicted elevation, $\bar{y}$ is the mean of the observed elevations, and n is the number of samples.

5) National-Scale Beach Topography Generation and Accuracy Evaluation: After optimization, the best-performing model from the four ML algorithms was applied to generate 10-m resolution DEMs for all sandy beaches across NZL. In GEE, we divided the entire study area into 31 blocks for processing [47]. We then processed each block separately, converting 13 feature images into machine-readable feature columns. These inputs were systematically integrated into the elevation prediction models. The predicted results were then inversely standardized using the normalization parameters derived from the training set. The block-scale results were mosaicked together to generate a comprehensive national-scale beach topography database for NZL.

To evaluate the accuracy of the generated DEM raster results, we selected ten beaches as validation areas. The

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TABLE VI
ESSENTIAL INFORMATION OF THE TEN BEACH VALIDATION AREAS, INCLUDING THEIR FIDS

Beach FID	Lidar DEM Coverage	Beach area ($10^4 \times m^2$)	NZL Peninsulas
638	0.91	42.49	North island
625	1.00	3.47	North island
623	1.00	4.22	North island
633	0.96	3.61	North island
614	0.99	20.05	North island
189	1.00	22.65	South island
411	0.94	17.09	South island
439	1.00	8.60	South island
420	1.00	8.11	South island
387	1.00	7.95	South island

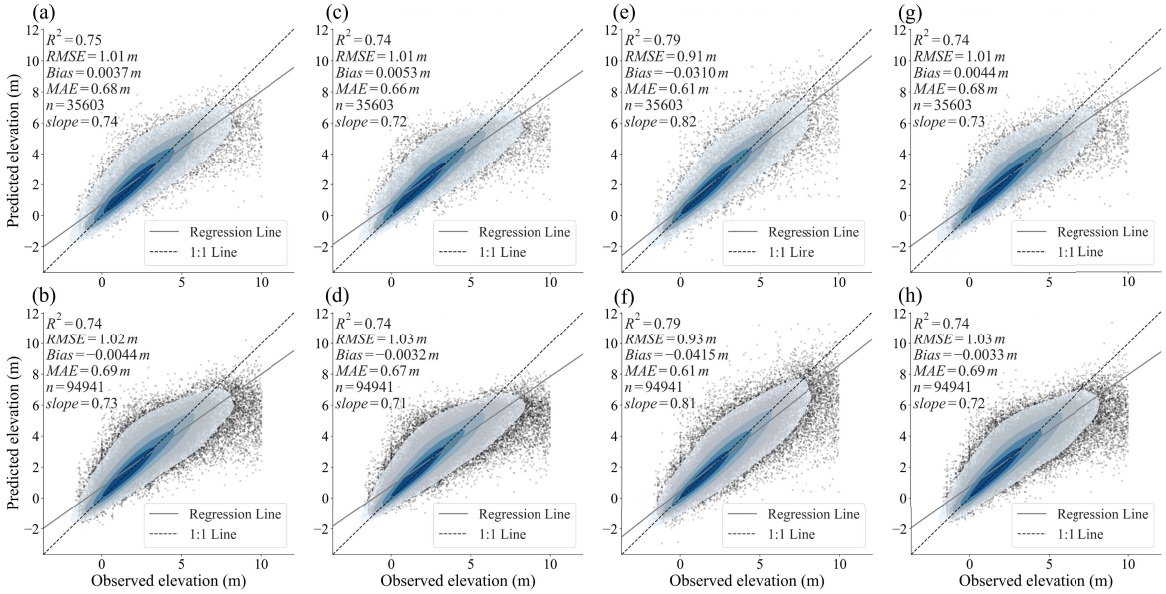


Fig. 5. Scatterplots depicting the relationship between observed and predicted elevations for the four models used in this study, with each model including results for both the test set and the validation set. Above the scatter is the point density layer, and the closer it is to dark blue, the higher the point density. Here, (a) and (b) are the XGB model's, (c) and (d) RF model's, (e) and (f) ANN model's, and (g) and (h) HGBR model's test and validation sets.

specific criteria are as follows: 1) each beach must have an area greater than 200 000 m^2; 2) LiDAR DEM coverage on the beach must exceed 0.9 (i.e., 90%) to ensure sufficient validation data; and 3) the beaches must be evenly distributed between NZL's North and South Islands, with five beaches randomly selected from each island to comprehensively evaluate elevation inversion accuracy across different regions. By comparing the original and predicted DEMs, we visualized the results using scatter plots and difference maps. The FIDs and other essential information of these ten beaches are summarized in Table VI.

IV. RESULTS

A. Model Performance and Analysis

Fig. 5 shows the scatter plots of predicted versus observed elevations for the four models. After retraining and retuning, the ANN now yields the best predictive accuracy on both the test and validation sets (test: R^2 = 0.79, RMSE = 0.91 m, MAE = 0.61 m, Bias = −0.0310 m, and slope = 0.82; validation: R^2 = 0.79, RMSE = 0.93 m, MAE = 0.61 m, Bias = −0.0415 m, and slope = 0.81). XGB, RF, and HGBR show similar, slightly lower performance (XGB test: R^2 = 0.75, RMSE = 1.01 m, MAE = 0.68 m, Bias = 0.0037 m, and slope = 0.74; validation: R^2 = 0.74, RMSE = 1.02 m, MAE = 0.69 m, Bias = −0.0044 m, and slope = 0.73. RF test: R^2 = 0.74, RMSE = 1.01 m, MAE = 0.66 m, Bias = 0.0053 m, and slope = 0.72; validation: R^2 = 0.74, RMSE = 1.03 m, MAE = 0.67 m, Bias = −0.0032 m, and slope = 0.71. HGBR test: R^2 = 0.74, RMSE = 1.01 m, MAE = 0.68 m, Bias = 0.0044 m, and slope = 0.73; and validation: R^2 = 0.74, RMSE = 1.03 m, MAE = 0.69 m, Bias = −0.0033 m, and slope = 0.72).

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TABLE VII
TRAINING SET ACCURACY AND BAYESIAN OPTIMIZATION TIME OF THE FOUR MODELS

Model type	R^2	RMSE (m)	MAE (m)	Bias (m)	Slope	Bayesian optimization time (min)
XGB	0.83	0.84	0.58	0.0001	0.79	5.85
RF	0.84	0.82	0.52	0.0015	0.77	99.17
ANN	0.86	0.76	0.52	-0.0336	0.85	853.75
HGBR	0.82	0.85	0.58	0.0000	0.78	43.70

Table VII reports the training-set statistics and the Bayesian-optimization durations for each model. On the training set, the ANN attains the highest goodness-of-fit (R^2 = 0.86, RMSE = 0.76 m, MAE = 0.52 m, Bias = −0.0336 m, and slope = 0.85), followed closely by RF (R^2 = 0.84, RMSE = 0.82 m, MAE = 0.52 m, Bias = 0.0015 m, and slope = 0.77), XGB (R^2 = 0.83, RMSE = 0.84 m, MAE = 0.58 m, Bias = 0.0001 m, and slope = 0.79), and HGBR (R^2 = 0.82, RMSE = 0.85 m, MAE = 0.58 m, Bias = 0.0000 m, and slope = 0.78). The Bayesian optimization time differs markedly among models (XGB ≈ 5.9 min; HGBR ≈ 43.7 min; RF ≈ 99.2 min; and ANN ≈ 853.8 min).

Taken together, the ANN provides the best balance of predictive accuracy and agreement with the 1:1 line (highest R^2, lowest RMSE/MAE on test and validation, and the steepest slope closest to unity), indicating superior generalization for this dataset. This improvement, however, comes at a substantial computational cost: ANN required an order of magnitude longer Bayesian optimization than the other methods. XGB and RF remain attractive when computational efficiency is a priority (XGB in particular is very fast to optimize), while HGBR offers a middle ground. Given the present study's emphasis on predictive accuracy and faithful recovery of elevation (including slope), the ANN was selected as the optimal model despite its substantially longer tuning time.

B. Model Interpretation

As shown in Fig. 6, the SHAP summary plot generated using KernelExplainer for the ANN model provides a global, model-centric explanation of how each input feature influences the predicted beach elevation. Features are ranked in descending order of importance according to their mean absolute SHAP values, which quantify each feature's average contribution magnitude to the model output. The sign of a SHAP value indicates the direction of the contribution (positive values push the prediction upward; negative values push it downward). Point color encodes the original feature magnitude (red = high and blue = low), and a wide color spread for a feature indicates that the feature assumes a broad range of values across the dataset and is therefore influential in many contexts. Fig. 6 shows that the two features with the largest SHAP ranges exert the greatest influence on model predictions. In particular, B8_50p (near-infrared, 50th percentile) is the single most important predictor: high B8_50p values (red points) are predominantly associated with positive SHAP values and therefore raise the predicted elevation. Conversely,

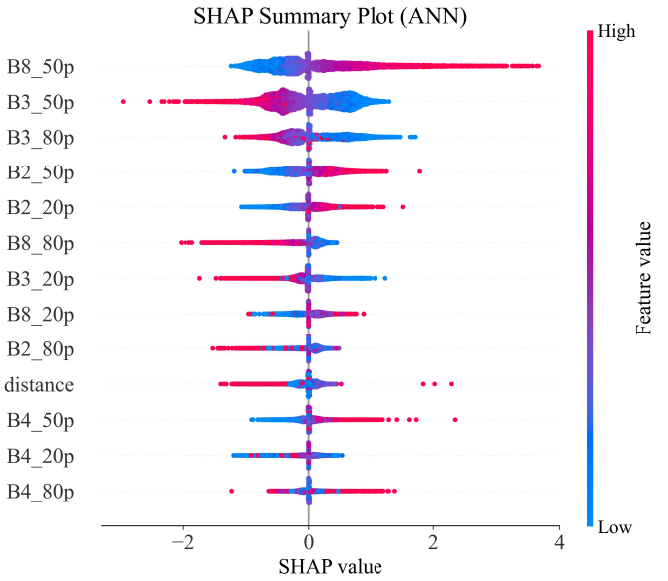


Fig. 6. SHAP summary plot of the ANN model. The plot illustrates the importance and impact of each feature on the model's predictions. Features are ranked by their mean absolute SHAP values (descending order), indicating their relative influence. Red and blue represent high and low feature values, respectively.

B3_50p (green, 50th percentile) is highly influential in the opposite direction: high B3_50p values correspond mainly to negative SHAP values and reduce the predicted elevation. Several other predictors contribute in both positive and negative directions depending on context; for example, B3_80p (green, 80th percentile) tends to increase predicted elevation when it is high, but its overall contribution is substantially smaller than that of B3_50p. Taken together, these patterns suggest a physically interpretable spectral–elevation mapping: higher reflectance in the green band (B3) tends to indicate moist or low-lying intertidal surfaces (hence lower elevation), whereas stronger NIR reflectance (B8) tends to indicate drier, coarser, or vegetated surfaces (e.g., dunes and backshore), which correspond to higher elevation. Thus, the model appears to rely primarily on mid-to-high percentile spectral signals (notably B8_50p and B3_50p) to discriminate among coastal geomorphic units (e.g., tidal flats versus dunes/backshore).

Fig. 7 presents the SHAP heatmap, which displays per-sample SHAP values for each feature across a representative subset of the dataset. In the heatmap, red indicates a strong positive contribution of a feature to the prediction for that

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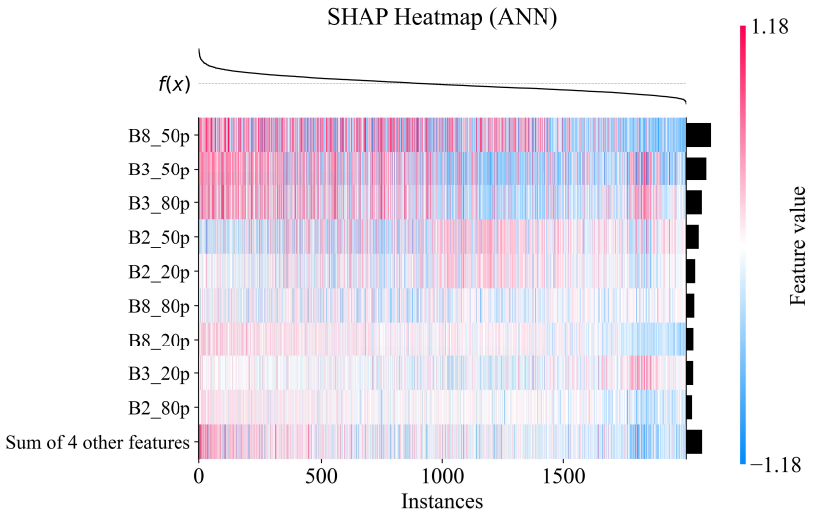


Fig. 7. SHAP heatmap of 2000 stratified samples. The heatmap visualizes the SHAP values of each feature across individual samples. Red and blue denote positive and negative contributions to predictions, respectively, while white indicates minimal influence. The upper black curve represents the overall trend of SHAP values from the first sample to the last sample. The black bar plot on the right side of the heatmap indicates the influence magnitude of each feature across all samples.

particular sample, blue indicates a strong negative contribution, and near-white indicates negligible influence. To ensure that the sampled subset preserves the elevation distribution of the full dataset and avoids bias toward specific elevation ranges, the 2000 samples used to generate the heatmap were drawn using stratified random sampling: the LiDAR DEM elevations were divided into 12 equal strata, and samples were randomly selected from each stratum in proportion to the stratum size. This stratified sampling strategy preserves the original elevation distribution and improves the interpretability of the heatmap. The heatmap confirms the patterns visible in the summary plot: high B8_50p values generally correspond to positive contributions, while low B8_50p values correspond to negative contributions. Moreover, the heatmap reveals variability in the direction and magnitude of feature contributions across individual samples, reflecting the nonlinear and context-dependent nature of spectral influences on elevation predictions.

To further quantify feature behavior at the instance level, Fig. 8 displays dependence plots for the six most important features. These plots show, for each feature, the relationship between the raw feature value (x-axis) and its SHAP value (y-axis), with points colored by a second feature to highlight interactions and context dependence. The dependence plots corroborate the summary and heatmap findings and additionally reveal modulation and nonlinearity: for example, the elevating effect of B8_50p is amplified when the red-band percentiles (B4_20p/B4_80p) are high, indicating a synergistic red–NIR signature for dry, elevated units; the depressing effect of B3_50p becomes stronger when B2 (blue) is also high, consistent with green+ue signatures of wet or tidal surfaces; and the 80th-percentile metrics (B3_80p, B8_80p) display nonlinear and regime-dependent behavior—their predictive significance varies with shoreline proximity and the baseline reflectance levels of other bands. These dependence plots, therefore, provide direct evidence of threshold-like responses and pairwise modulation that explain the joint regimes observed in the heatmap.

In summary, the combined use of the SHAP summary plot, heatmap (based on stratified sampling), and dependence plots demonstrates that the spectral–elevation relationships learned by the ANN are physically plausible and interpretable: B8_50p and B3_50p act as the primary predictive axes, while high-percentile indicators (80p) show context dependence.

C. National-Scale Beach Topography

The accuracy was assessed using the same metrics: R^2, RMSE, MAE, and Bias. The scatter distribution plots for the ten beaches are shown in Fig. 9. Overall, the accuracy varies across the ten beaches, with the highest accuracy observed at "FID 439" (R^2 = 0.88, RMSE = 0.47 m, MAE = 0.31 m, Bias = 0.0556 m, and slope = 0.96) and the lowest at "FID 638" (R^2 = 0.75, RMSE = 0.93 m, MAE = 0.67 m, Bias = −0.1000 m, and slope = 0.77). Despite some regional variations, the overall accuracy remains satisfactory (with an overall R^2 of 0.86 and RMSE of 0.75 m across the ten beaches), closely approximating the performance of the ANN model on the validation and test sets, indicating strong model generalizability. The raster maps of the elevation prediction results for the ten beaches are presented in Fig. 10. From the results, the spatial distribution of elevation follows the expected pattern, with elevation values gradually increasing from the coastline inland. In addition, we generated images of spatial distribution of residuals by subtracting the LiDAR DEM (−2 to 10 m) from our study's DEM (as shown in Fig. 11). The results demonstrate that for all ten beaches, the majority of pixels cluster near yellow tones (indicating values close to zero), confirming only minor discrepancies exist between our derived DEM and the original DEM data. In conclusion, this approach can be effectively utilized for national-scale beach topographic mapping of NZL at a 10-m resolution DEM.

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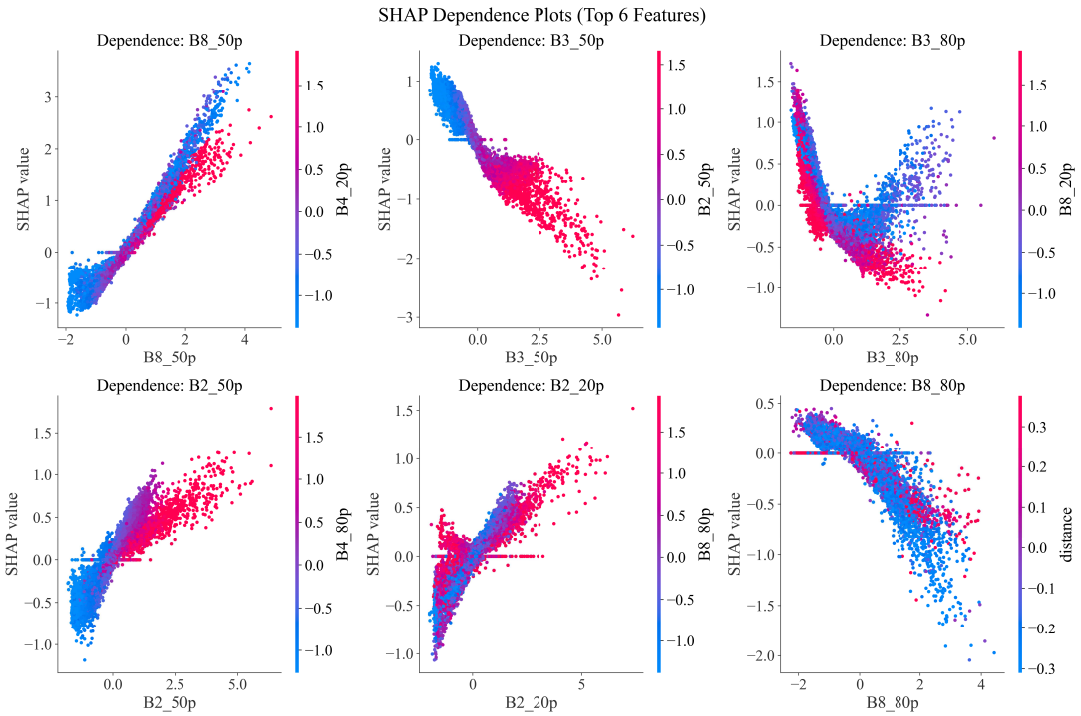


Fig. 8. SHAP dependence plots for the top six most important features. In each dependence plot, the *x*-axis represents the feature value, while the *y*-axis shows the corresponding SHAP value (indicating the feature's contribution to the model output). The point colors represent values of an interactive feature, with the color gradient from blue to red indicating increasing feature values.

V. Discussion

A. Stratified Accuracy Analysis and Tidal Effects

Fig. 12 summarizes model accuracy (RMSE) and systematic error (Bias) stratified by elevation intervals alongside the sample counts per bin. Two salient patterns emerge from these stratified diagnostics. First, the lowest elevation bin (−2 to −1 m) exhibits relatively large RMSE and a pronounced positive bias. Second, prediction errors (both RMSE and negative bias) increase markedly at the upper end of the elevation spectrum (≈ 5 m and above), despite some elevation bins in this range containing nonnegligible sample numbers.

These patterns can be explained by a combination of i) sampling availability controlled by LiDAR acquisition timing (tidal-state effects), ii) the spectral sensitivity of Sentinel-2-derived predictors to different substrate/moisture conditions, and iii) the changing surface composition (e.g., vegetation and cobble/rock) across the cross-shore transect.

First, the poor accuracy in the lowest elevation class is directly linked to tidal-state sampling of the airborne LiDAR. The LiDAR surveys used here were acquired at discrete times and therefore sample the coastal surface at specific tidal stages; consequently, the lowest intertidal surfaces are often undersampled or entirely absent from the LiDAR footprint (see Fig. 2(a) and the data acquisition discussion). This underrepresentation reduces the quantity (and sometimes diversity) of reliable training labels at the lowest elevations, which in turn increases prediction uncertainty and produces the larger RMSE and variable bias observed in the −2 to −1 m bin. Similar tidal/temporal sampling issues affecting LiDAR-based coastal datasets have been noted in prior studies and motivate the need for auxiliary lower elevation observations when the objective is a continuous land–sea terrain model [17], [20].

Second, the spectral predictors derived from Sentinel-2 composites (percentile-based blue/green/red/NIR bands) are intrinsically most sensitive to differences in surface wetness and substrate texture, conditions that are pronounced in the intertidal and wet sand/mudflat domains. In short, variability in B2/B3/B8 percentiles maps well to changes in water content and graininess common in tidal flats, so the model can exploit strong spectral–elevation coupling in the low-elevation regime (helping predictive skill there when training labels are available). By contrast, the landward/backshore and dune domains often include vegetation, coarse gravels, or rocky exposures whose spectral signatures are less tightly coupled with elevation per se. Therefore, even where some high-elevation bins contain many samples, the spectral predictors provide weaker elevation information for these surface types; this spectral insensitivity leads to the larger RMSE and the systematic underestimation trend seen above ∼5 m. This interpretation is consistent with the SHAP analysis (see Section V-B), which shows that band-percentile predictors are most informative for distinguishing wet/low versus dry/high surfaces, but are less effective at discriminating among vegetated or rocky high-elevation units.

Finally, sample scarcity compounds these effects. Bins with small sample counts (either at extreme low or extreme high elevations) show the largest spread in RMSE and bias in Fig. 12. Limited training examples reduce the model's ability to learn robust mapping functions in those regimes

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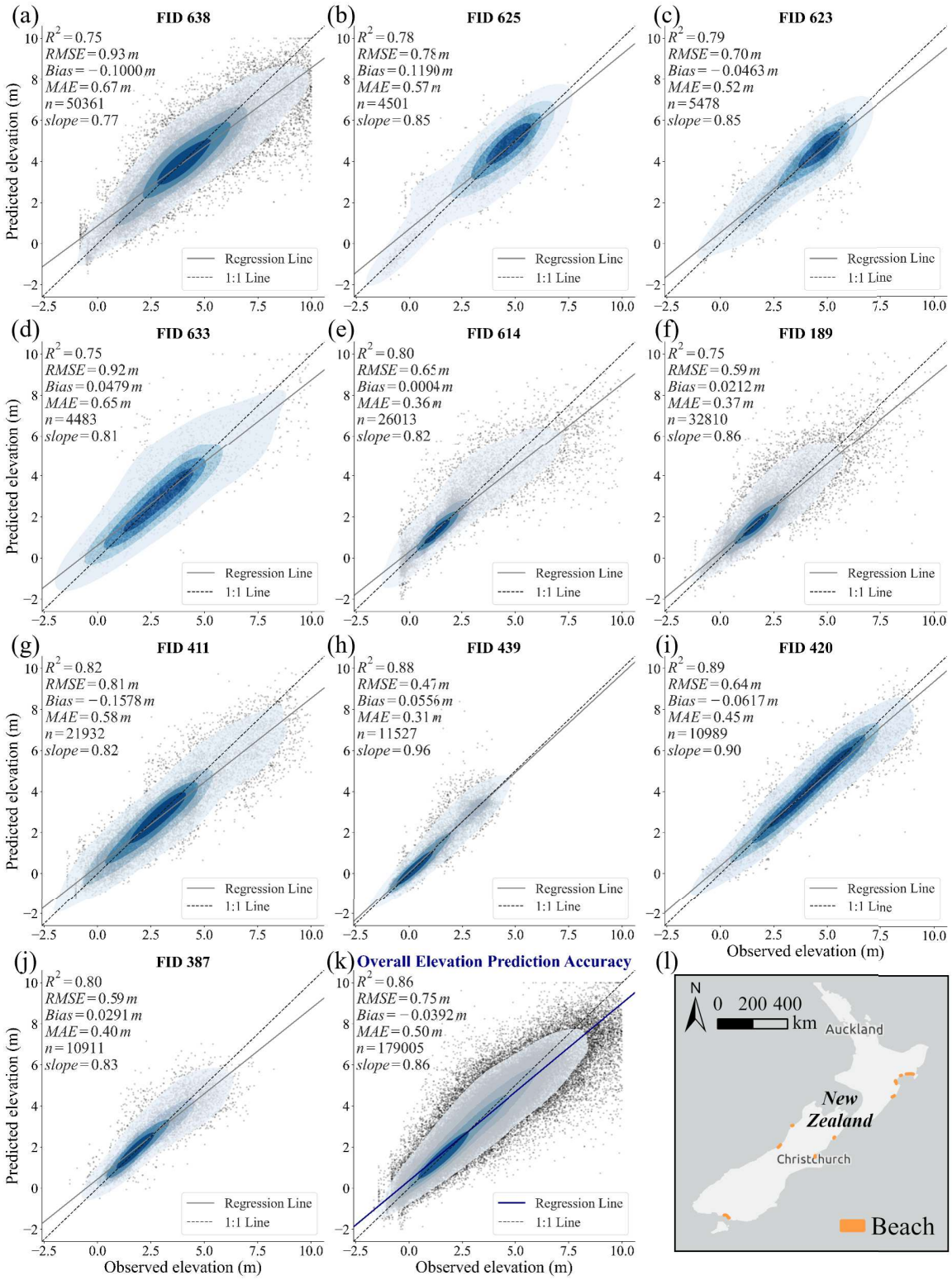


Fig. 9. (a)–(j) Scatter relationship between observed and predicted elevations across ten beach areas. (k) Overall accuracy of ten beaches. (l) Approximate locations of the ten beach surfaces.

and increase sensitivity to outliers. Where both spectral information is weak (e.g., vegetated backshore) and samples are few (e.g., rare high dune crests), prediction errors amplify—explaining why some high-elevation bins with relatively few points still show poor accuracy.

B. Analysis of Extended Beach Topography

By calculating the total area of valid pixels in both the original DEM and the predicted DEM, we assessed the changes in DEM coverage for all beach regions across NZL.

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Fig. 10. 10-m resolution elevation prediction results for selected areas of ten beaches. The FID values for the ten beaches from (a)–(j) are: 638, 625, 623, 633, 614, 189, 411, 439, 420, and 387. (k) Approximate locations of the ten beach surfaces.

The area covered by the DEM derived from airborne LiDAR data was 39.83 km^2, while the area covered by the DEM generated in this study was 153.98 km^2. This represents a significant increase of 114.15 km^2 (increasing the area

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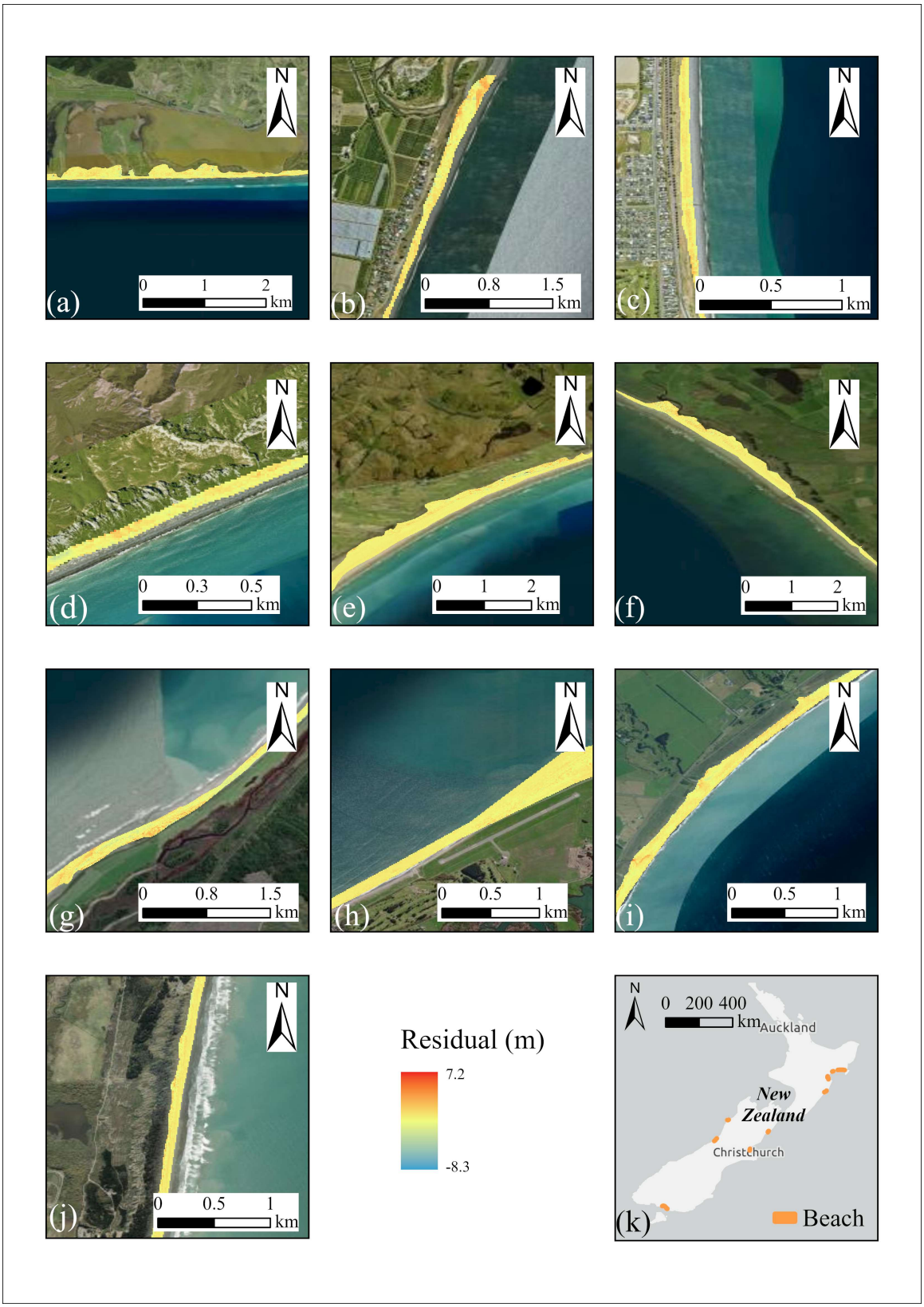


Fig. 11. Spatial distribution of residuals (predicted DEM minus LiDAR DEM) between the predicted DEM and LiDAR DEM across ten beach areas, with values ranging from −2 to 10 m. The FID values for the ten beaches from (a)–(j) are: 638, 625, 623, 633, 614, 189, 411, 439, 420, and 387. (k) Approximate locations of the ten beach surfaces.

by approximately 286.6%) in the DEM coverage area. In addition, the original LiDAR DEM data covered a total of 359 beach surfaces, while the DEM data obtained in this study covered 1608 beach surfaces. DEM data were newly added

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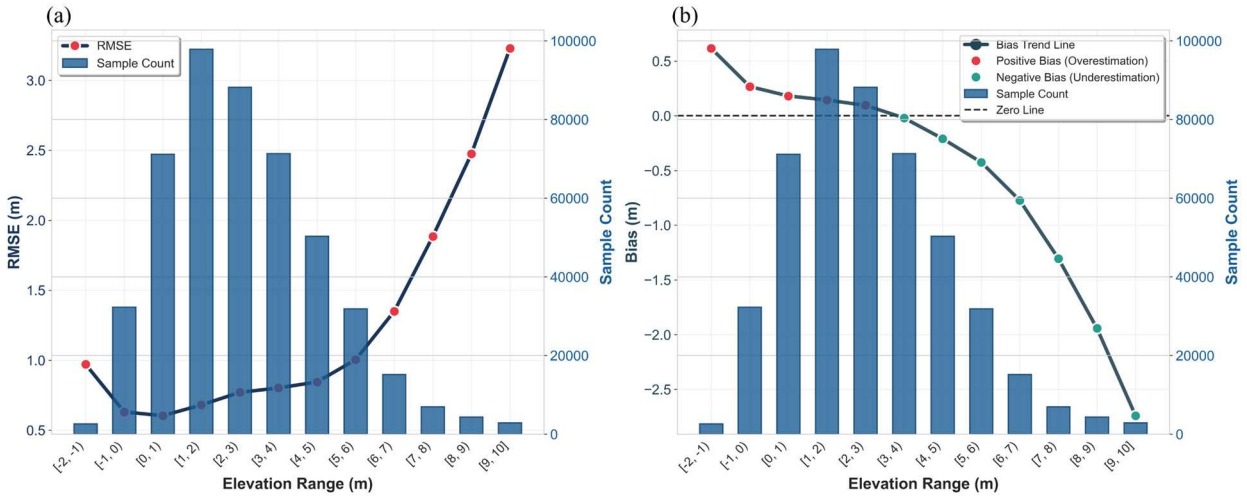


Fig. 12. Stratified RMSE, Bias, and sample counts by elevation. (a) RMSE (line) and sample count (bars) per elevation bin. (b) Bias (line) and sample count (bars).

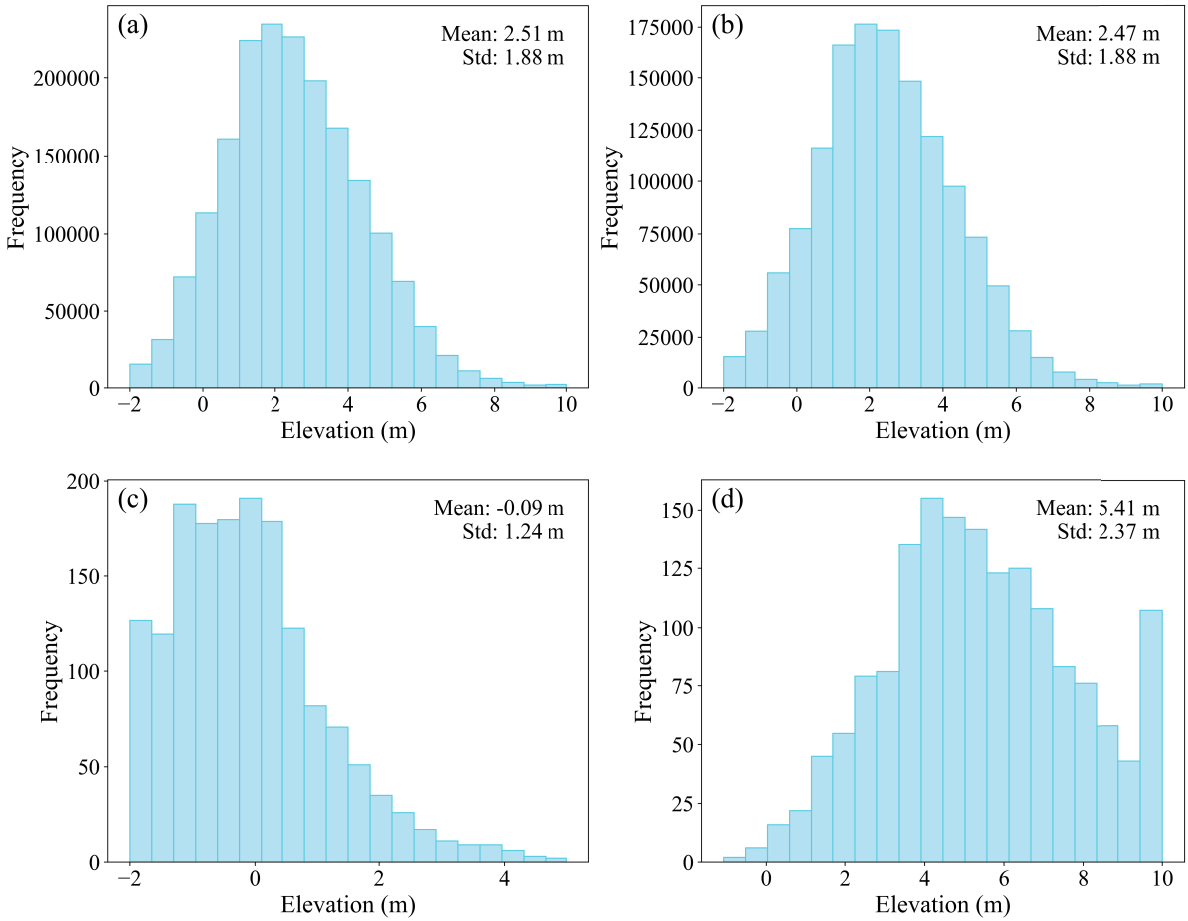


Fig. 13. Elevation distributions for the study area's predicted DEM and the newly added region: (a) overall histogram of predicted DEM elevations; (b) overall histogram of elevations in the newly added area; (c) per-beach minimum elevations; and (d) per-beach maximum elevations.

for 1249 complete beaches (an increase of approximately 347.9%).

Fig. 13 shows the elevation distribution (total elevation, minimum, and maximum values within each beach) of the research results (predicted DEM) in our study, as well as the elevation distribution of the newly added area. Fig. 13(a) and (b) displays the elevation distribution of the predicted DEM and that of the newly added area. The similar mean

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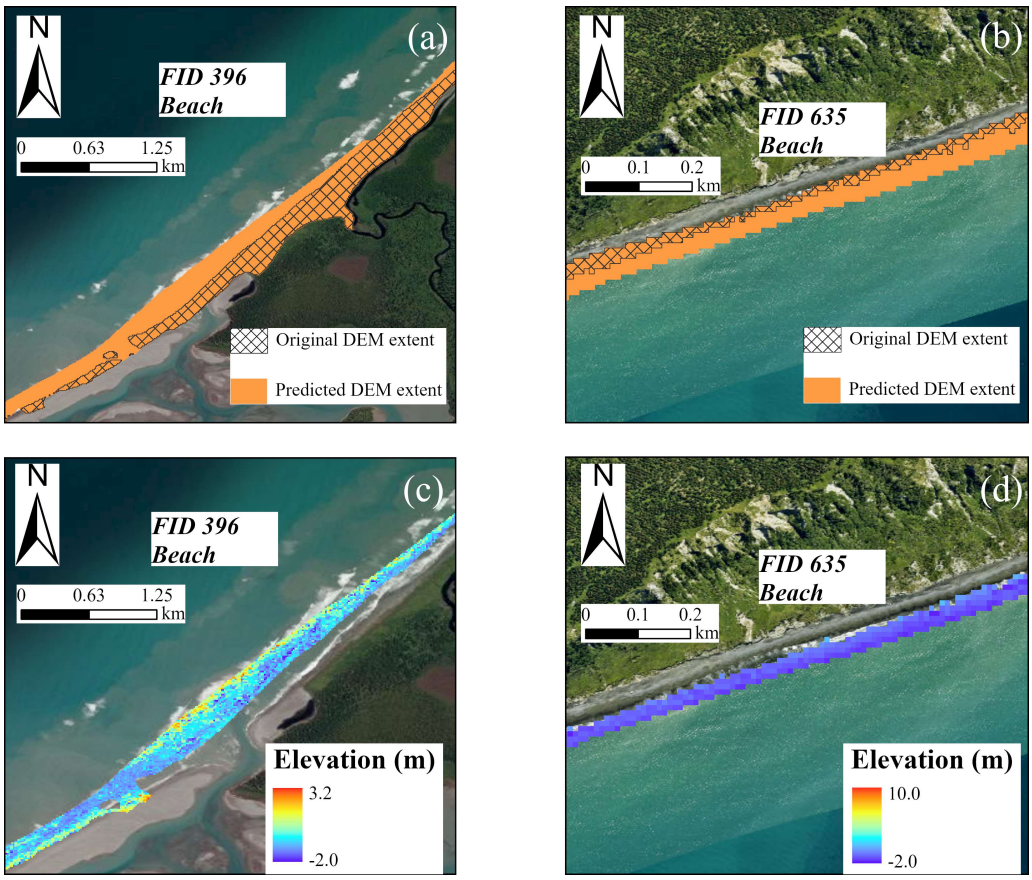


Fig. 14. Added areas corresponding to beaches with FIDs 396 and 635: (a) and (b) boundary extents of the original DEM and the predicted DEM obtained in this study and (c) and (d) elevation distribution of the newly added beach areas.

values and standard deviations (2.51 versus 2.47 m and 1.88 versus 1.88 m, respectively) indicate that the newly added area constitutes a large portion of the total region and is predominantly characterized by low elevation values. Fig. 13(c) and (d) shows the distribution of the maximum and minimum values of the newly added area on each beach. The distribution of the minimum values reveals extensive interpolation of low-elevation areas, while the distribution of the maximum values—with a mean of 5.41 m—reflects the inclusion of large newly added beach areas such as Stewart Island, which contain extensive higher elevation regions.

As shown in Fig. 14, notable raster coverage expansion is also evident in the beach areas with FIDs 396 and 635 [see Fig. 14(a) and (b)], accompanied by distinct interpolation of low elevation values in their coastal zones [see Fig. 14(c) and (d)].

Subsequently, comparison of the two DEM raster images revealed a substantial expansion of beach coverage in the generated DEM from our study. Notably, increased coverage was observed not only in remote Stewart Island (see Fig. 15), but also across multiple large beach regions, indicating a significant overall increase in beach areas.

The expansion of DEM coverage at the beach scale demonstrates the dual advantages of our product dataset: high accuracy and enhanced data coverage. First, the predicted DEM maintains a high level of precision, as evidenced by the validation metrics (e.g., R^2, RMSE, MAE, and Bias), ensuring reliable elevation data for coastal analysis. Second, the significant increase in coverage area addresses critical gaps in existing datasets, particularly in regions where airborne LiDAR data were previously unavailable or limited. This expanded coverage is especially valuable for remote or understudied beaches, such as those on Stewart Island, where traditional data collection methods are often logistically challenging and costly.

Extending DEM coverage while preserving accuracy provides a more comprehensive and consistent dataset for beach-scale analysis. This is particularly advantageous for applications such as coastal erosion monitoring, sediment transport studies, and habitat mapping, where detailed and spatially extensive elevation data are essential. Furthermore, the improved dataset supports better-informed decision-making in coastal management and risk assessment, particularly in areas vulnerable to sea-level rise and extreme weather events.

C. Global Scalability and Alternative Data Sources

This study integrated airborne LiDAR and Sentinel-2 optical imagery to produce a 10-m DEM for beaches along NZL's coastline at a national scale. However, the high cost and limited coverage of LiDAR data pose significant challenges, particularly on a global scale, where many regions lack high-precision LiDAR data. Therefore, exploring alternative elevation data sources is crucial for global beach topographic mapping.

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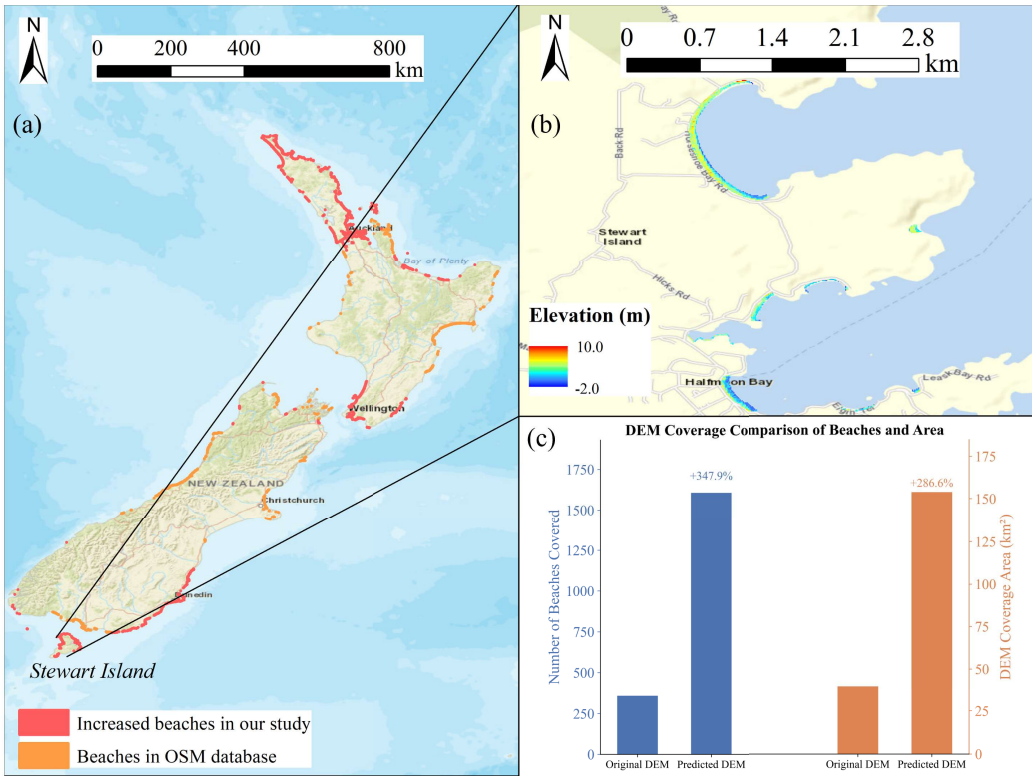


Fig. 15. Spatial visualization of DEM coverage enhancement: (a) beach distributions from OSM (orange) and newly acquired DEM coverage (red); (b) coverage expansion on Stewart Island; and (c) changes in beach counts and total coverage area.

Fortunately, ICESat-2 (Ice, Cloud, and Land Elevation Satellite-2) data emerge as a promising alternative [63]. ICESat-2 is equipped with the Advanced Topographic Laser Altimeter System (ATLAS), which provides global-scale high-precision elevation measurements [63]. Compared to airborne LiDAR data, ICESat-2 offers several advantages: First, its global coverage enables the provision of elevation data for regions lacking airborne LiDAR data [64]. Second, ICESat-2 data exhibit high vertical accuracy (typically at the centimeter level), enabling accurate beach topographic mapping. Note that some existing studies have demonstrated the applicability of ICESat-2 data in beach topography inversion [17], [65].

Although this framework was developed and validated for NZL, the approach is inherently transferable to other geographic contexts. Where airborne LiDAR is unavailable, alternative elevation sources can be selected according to local data availability and application requirements. In addition to ICESat-2, candidate datasets include TanDEM-X and other interferometric DEMs, high-resolution satellite stereo (e.g., Pléiades), aerial stereo imagery, UAS-SfM photogrammetric DEMs, and multibeam bathymetry for nearshore areas. Publicly available coastal DEM compilations (e.g., DeltaDTM) can also be used as supplementary inputs or benchmarks for regional modeling and error assessment [38]. The pragmatic selection and combination of these sources should be guided by the target region's geomorphology, the desired vertical accuracy, and data accessibility or cost considerations. Moreover, we highlight that this approach exhibits great potential in producing high-precision DEMs for beaches on a global scale, and is able to provide valuable support for coastal zone management and facilitate sustainable resource utilization.

D. Future Considerations

To construct a spatially continuous and physically consistent coastal terrain suitable for sea-level rise assessments and storm-surge modeling, it is necessary to augment the elevation sampling both seaward into the lowest intertidal and shallow subtidal zones and landward across dunes and backshore areas. The airborne LiDAR data employed in this study were acquired at discrete survey times and therefore represent instantaneous tidal states; as a consequence, the lowest intertidal surfaces are often undersampled or absent from the LiDAR footprint. Likewise, certain landward geomorphic elements (e.g., dune crests and vegetated backshore) may be under-represented in the LiDAR-based training set. These gaps produce an imbalanced elevation distribution in the training data and increase the likelihood of model extrapolation and error at the extremes of the elevation spectrum.

A pragmatic pathway toward an integrated coastal terrain is to combine complementary remote sensing products that collectively span the full cross-shore elevation range. For the nearshore domain, satellite altimetry (notably ICESat-2) and validated satellite-derived bathymetry (SDB) products offer pointwise or spatially continuous elevation/bathymetric information in regions lacking airborne bathymetric surveys, thereby helping to populate the lower end of the elevation distribution [32]. For the landward domain,

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high-resolution stereo photogrammetry—derived from submeter satellite stereo pairs, aerial stereo imagery, or UAS-SfM surveys—can provide dense coverage of dunes and backshore zones that are often undersampled by regional LiDAR acquisitions [53]. When integrated appropriately, these auxiliary datasets extend the elevation range of training samples and reduce the need for model extrapolation at both low and high elevations, improving model robustness across geomorphologically diverse coastlines.

To enhance applicability and transferability across regions, several strategies should be considered. First, feature diversity and physically informed predictors can improve model generalization: in addition to the band-percentile composites used here (20p/50p/80p), incorporating ancillary predictors—such as tidal-state proxies, wave exposure indices, substrate/land-cover maps, and terrain attributes (e.g., slope and curvature)—can strengthen the spectral–topographic linkage. Second, when multiple auxiliary elevation datasets are available, deriving consensus or uncertainty-aware features (for example, ensemble mean and variance of co-registered DEMs) can help the model to discount noisy inputs and generalize more effectively. Third, transfer learning and domain adaptation workflows offer a practical route to regional deployment: a model pretrained in a data-rich domain (such as NZL) can be fine-tuned using a modest number of local reference samples, and domain-alignment techniques (e.g., feature alignment or adversarial adaptation) can mitigate covariate shifts between sensors or surface types. Methods that explicitly incorporate topographic priors or physical constraints into learning architectures have also shown promise for DEM void-filling and reconstruction tasks and may further reduce artifacts when transferring models between regions [34].

An additional, complementary measure to improve elevation inversion is to refine beach boundary delineation. Imprecise or inconsistent beach masks introduce mixed or nonrepresentative spectral samples into both training and validation sets, which degrades model performance and complicates error attribution. Recent research demonstrates that Sentinel-2-based classification combined with object-level processing can produce accurate, high-resolution sandy-beach maps suitable for large-area applications [66]. Integrating such beach delineation methods—leveraging temporal composites, spectral indices, and object-based post-processing—would i) reduce contamination by nonbeach land cover, ii) enable stratified sampling that targets pure geomorphic classes (e.g., tidal flat, foreshore, and dune), and iii) improve spatial consistency and interpretability of ML-derived DEMs. We therefore recommend that future workflows incorporate automated or semiautomated beach delineation and report boundary uncertainty metrics alongside DEM accuracy statistics.

Several practical caveats warrant emphasis. Temporal mismatches, tidal-state dependencies, and vertical datum inconsistencies among heterogeneous products can introduce systematic biases; therefore, multitemporal sampling, tidal normalization, rigorous co-registration, and bias correction are recommended when assembling auxiliary datasets. Characterizing and propagating input uncertainties through the modeling chain is critical to quantify the integrated DEM's fitness for purpose. Finally, independent validation—using reserved LiDAR segments, in situ surveys, or other hold-out datasets—remains indispensable to quantify accuracy and uncertainty before operational application.

VI. Conclusion

This study has developed and validated a novel ML-based framework for reconstructing high-resolution (10 m) beach topography at a national scale in NZL by integrating airborne LiDAR, Sentinel-2 multispectral imagery, and geometric features. The main conclusions are as follows.

1) The ANN model demonstrated superior performance in predicting beach elevation, achieving an R^2 of 0.79 and an RMSE of 0.93 m on independent validation data, outperforming other widely used ML models (XGB, RF, and hist gradient boosting). Model interpretation via SHAP analysis confirmed that spectral features—particularly the 50th percentile of near-infrared (B8) and green (B3) bands—were the most influential predictors, exhibiting physically interpretable relationships with coastal geomorphic units.
2) The proposed framework significantly expanded DEM coverage across NZL's sandy beaches. The predicted DEM increased the total mapped beach area by 286.6% (114.15 km^2), extending coverage to 1608 beaches—including 1249 previously unmapped beaches—with particular improvements in remote regions such as Stewart Island.
3) Extensive validation across ten representative beaches confirmed the robustness and generalizability of the method, with site-specific R^2 values ranging from 0.75 to 0.88 and RMSE values between 0.47 and 0.93 m.

This study provides the first national-scale beach-specific topographic dataset for NZL, offering a scalable and transferable solution that combines broad spatial coverage with high spatial resolution. The proposed approach effectively overcomes the limitations of traditional LiDAR-based surveys—such as tidal constraints, spatial gaps, and high costs—and establishes a new paradigm for large-scale coastal elevation mapping. The resulting DEM product supports a wide range of applications, including coastal hazard assessment, habitat monitoring, and climate adaptation planning, thereby contributing to sustainable coastal zone management and resilience building under climate change.

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